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Three-Level ANSI/SPARC Architecture

CSA0503 - Database Management Systems

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Abstract

- ANSI/SPARC is a standard architecture for organizing database systems.
- It divides a database into **three levels**: External, Conceptual, and Internal.
- The **External Level** provides customized views for different users.
- The **Conceptual Level** defines the overall logical structure of the database.
- The **Internal Level** describes how data is physically stored in storage devices.

Introduction

What is ANSI/SPARC Architecture?

- Introduced to organize database systems into different levels.
- Separates user views from physical storage.
- Provides **data independence**, **security**, and **flexibility**.

Objectives

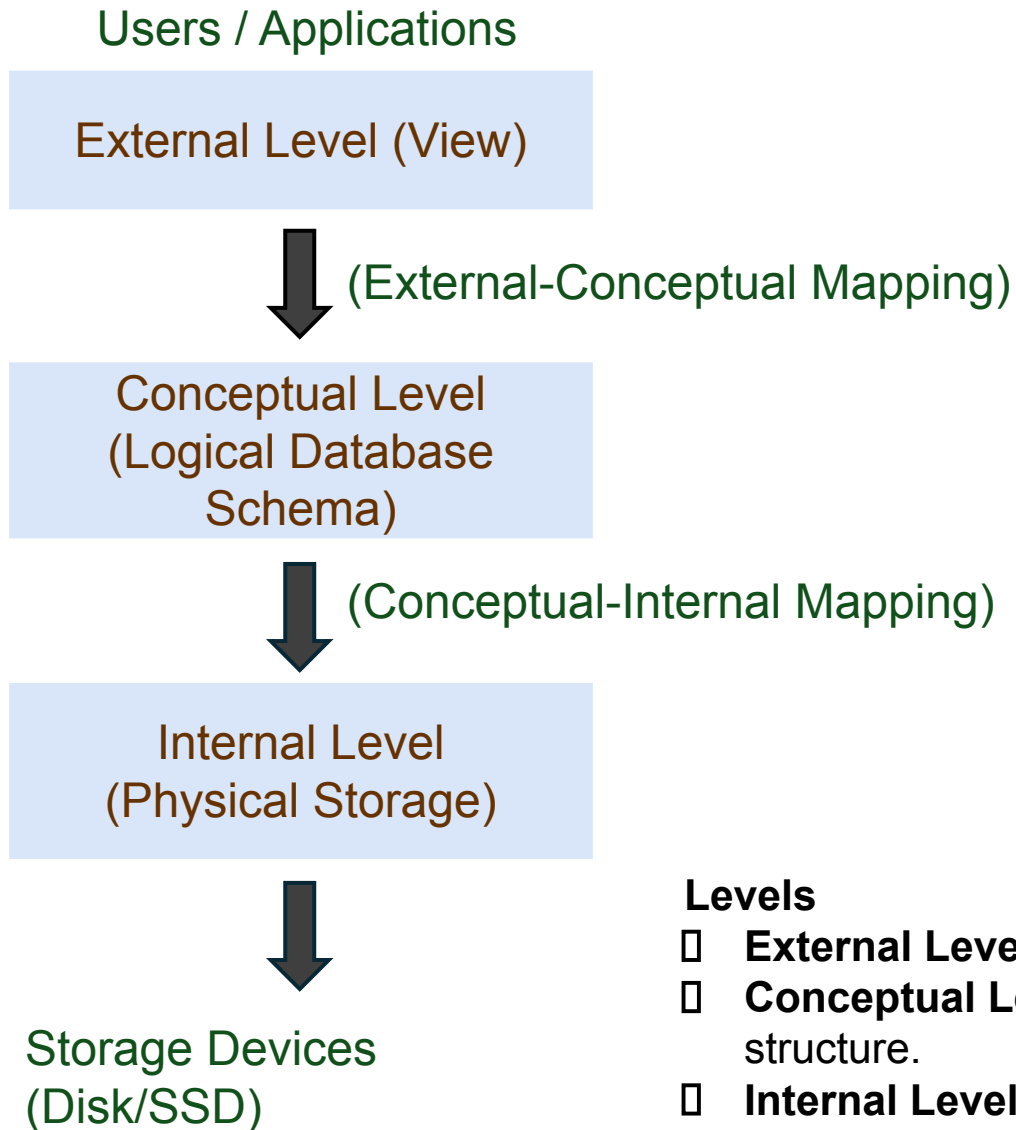
- Understand the three levels.
- Learn how data flows between levels.
- Explore practical examples.



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Three-Level ANSI/SPARC Architecture



Levels

- **External Level** – User-specific views.
- **Conceptual Level** – Overall logical structure.
- **Internal Level** – Physical storage details.



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External Level

Definition

- The **External Level** provides customized views of the database for different users.

Features

- Multiple user views.
- Hides unnecessary information.
- Improves security.

Example

Student View

- Name
- Roll Number
- Marks

Faculty View

- Student Name
- Marks
- Attendance

Each user sees only the required information.



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Conceptual Level

Definition

- Represents the complete logical structure of the database.

Includes

- Tables
- Relationships
- Constraints
- Data types

Example

College Database

Contains:

- **Student** – Student ID, Name, Department
- **Course** – Course ID, Course Name, Credits
- **Faculty** – Faculty ID, Name, Department
- **Enrolment** – Links students to the courses they have enrolled in

The conceptual level defines how data is logically connected.



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Internal Level



Definition

- Describes how data is physically stored.

Includes

- Files
- Indexes
- Storage blocks
- Compression

Example

Disk

- | -- Student.dat
- | -- Course.dat
- | -- Enrollment.dat
- | -- Index Files

Users never interact directly with this level.

Advantages

- Better security
- Easier maintenance
- Multiple user views
- Data independence
- Efficient database management



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Conclusion



- ANSI/SPARC architecture separates the database into **External, Conceptual,** and **Internal** levels.
- It provides **data independence, security,** and **efficient database management.**
- This architecture is widely used in modern Database Management Systems (DBMS).

References

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*Thank
you!*